STUDENTS LEARNING MATERIALS

Population aging and intergenerational relationships

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Course description

The course examines the challenges and opportunities of population aging and how it affects the distribution of resources and public policies' implementation. It will address topics such as population aging and changing age structures (emphasizing that aging is not only about older people), aging population trends, the health and well-being of older adults, the economic security and social protection of older adults, and the impact of aging on the workforce, employment and care. The course also focuses on the transfer of resources between generations, exploring how public policies can, and should, ensure a fair distribution of resources between younger and older populations.

Contents

1. What is population aging? Definition and measurement, trends, patterns, and components.
2. Economic implications of population aging: Changes in the labor market structure, pension systems, and sustainability of welfare states. The concept of the demographic dividend.
3. Population aging and health: Implications of the increase of life expectancies and the number of adults that will need a disease treatment and their relationship with health costs.
4. Social implications of population aging: Explore social care demand and provision, as well as intergenerational relations of care (for adults and grandchildren).

Learning Highlights

The course will offer a broad overview of the challenges and opportunities of population aging. It will help students to develop critical thinking on the topic, enabling them to challenge alarmist and ageist messages, usually coming from the media. It will also allow them to thoughtfully assess the importance of effective planning and informed decision-making in shaping public policies. Students will understand the importance of addressing population aging, from an intergenerational solidarity perspective, not as an issue affecting only older populations, but all age groups.

1 Population Aging

1.1. Definition of Aging

Population aging is different from **individual aging**, although terms are sometimes being confused because they use the same word (aging). Individual aging refers to the biological, psychological, and social changes that occur as people age, and it represents all the imaginary about the "aging" word: living longer, getting older, wiser and frailer. Meanwhile, population aging refers to the age structure (population pyramid) and it applies to the increase in the relative proportion of older age groups in the population. Although it is affected by the longer lives of individuals, it could also occur without increases in the number of years lived, caused by the shrinking base of the population pyramid.

During the **demographic transition**, fertility and mortality rates went down, causing a change in the population age structure, with a lower proportion of children, a temporary higher proportion of working-age population, and a later increase of older individuals. Population aging refers specifically to the growing share of older individuals. Although fertility, mortality, and migration can contribute to changes in the age structure of populations, the main responsible for this trend in the recent decades has been the decrease in birth rates.

Most countries in the world are facing population aging, which is a unique phenomenon, as has never happened before in human history. At the same time, pathways to population aging continue to be extremely diverse around the world, and not linear. Fertility declines have gone at very different speeds, as well as decreases in adult mortality levels, which results in very diverse population aging processes.

1.2. Components of Population Aging

The age structure of the population is defined by three components: birth rates, mortality rates, and migration. However, birth rates, highly influenced by fertility rates, have been traditionally argued to be the main component defining the population pyramid structure. In the near future, though, and due to the further stability of lower fertility rates, mortality levels and migration flows will become more important to draw population age structures.

Several demographers have delved into the demographic factors behind population aging, and the conclusions are very diverse. In Scott and Canudas-Romo (2024) they show that both births and survivorship define the growth in population aging, although countries show very different trends, and they highlight the importance of migration for the working age population. At the same time a work by De Santis and Salinari (2023) confirms that it is mainly mortality profiles that have shaped population age structures. At the European level, the work from Kashnitsky et al. (2017) evidenced the diversity of population aging trends in European regions. In this study they show the growing importance of the mortality and migration components in projections of population aging as fertility trends are forced to converge.

To conclude, there is an open debate about which are the prominent factors defining population aging trends. Their further definition is extremely important to understand how policies might influence each of the demographic components and, therefore, contribute better to reducing generational dependency.

1.3. Importance of Measuring Population Aging

Population aging is challenging in many aspects. The most striking ones relate to pension schemes, healthcare systems, and long-term care services. All of them rely heavily on the organization of welfare systems and intergenerational transfers that come from working-age populations. However, these challenges depend on the way dependents and contributors are defined. Therefore, how aging indicators are defined would be crucial to send a alarmist message.

Moreover, population aging comes with some benefits that need to be considered. Older adults today are healthier, better educated, and more active than in past generations. This means that many of them, once they are retired, may contribute to society through volunteering, caregiving, and sharing knowledge and experience. This is emphasized by the fact that with shrinking fertility rates, they can devote more time to their fewer adult children and grandchildren. However, again, all these benefits are subject to a healthier and more active longevity, which should be ensured through policy investments at the individual and neighborhood level.

Nevertheless, population aging would be a temporary stage, that would be reduced in the future if fertility and mortality rates stabilize. However, fertility rates under 2.1 (replacement fertility) will necessarily lead to declines in population. Lee et al. (2014) considered all economic profiles (private and public), using data from 40 different countries, and showed that higher fertility rates, above the replacement level, would clearly benefit government budgets. Fertility rates near replacement and population stability are more advantageous for family standards of living when it is considered the effects on private economic profiles from families, jointly to government transfers. Finally, fertility moderately below replacement and population decline would increase standards of living when human capital investments are done on the growing labor force.

1.4. Population Aging Indicators

There are many ways of measuring population aging, but its indicators have different policy implications. The United Nations has historically used a simple demographic measure, only capturing shares of age groups in the population. These indicators refer to the dependency levels, defining young and older individuals as dependents:

$$Total\ Dependency\ Ratio = Old\ age\ dependency\ ratio + Youth\ depdency\ ratio$$

$$Old\ age\ dependency\ ratio\ (OADR) = \frac{Population\ 65\ years\ old\ and +}{Population\ 15 - 64\ years\ old} x\ 100$$

$$Youth\ dependency\ ratio\ (YDR) = \frac{Population\ 0 - 14\ years\ old}{Population\ 15 - 64\ years\ old} x\ 100$$

$$Support\ Ratio\ (SR) = \frac{Population\ 15 - 64\ years\ old}{Population\ 0 - 14 + Population\ 65 +} x\ 100$$

This is a purely demographic measure that is extensively used to increase concern about health and economic issues. However, it does not tell us much about the level of dependency of the population, as dependency relies on the levels of activity and production, which might have changed over the years. We are now much healthier than before, and we are arriving at older ages more active. In fact, as life expectancy has increased, age, as a measure to define aging, has also changed.

"The concept of age has become more complicated because life expectancy has increased and people at each age have had progressively more remaining years of life. As people modified their behavior to reflect these changes, 40-year-olds began to act more like 30-year-olds had acted in the past." (Sanderson & Scherbov, 2008).

These authors propose that instead of using the chronological age, we should use the *prospective age*, which would be the number of remaining years of life expectancy. For example, in Table 1, we have the remaining life expectancy of French Women at age 30 in 1952 and 2005. The increase in life expectancy is clearly reflected. So, French women have the same remaining life expectancy at age 30 in 1952 than French women at age 40 in 2005. This follows the reflection regarding the fact that nowadays 40s are the past 30s. Although this concept is assuming that behaviors are strongly defined by the number of years left, it is true that older people are healthier, more active, and more engaged in social activities. If we use this prospective age to define old-age, the old-age indicator would consider the increase in life expectancy as part of the definition.

To obtain a Prospective Old-Age Dependency Ratio (POADR), it is necessary, first, to define the prospective age at which we decide an old person can be considered as such. In Sanderson & Scherbov (2007; 2008), for example, they use a remaining life expectancy of 15 years to define old-age.

Table 1. Remaining Life expectancy of French women at different ages and years

Year	Chronological age	Remaining Life expectancy (Prospective age)
1952	30	44.7
2005	30	54.4
2005	40	44.7

Source: Human Mortality Database used from Sanderson & Scherbov (2008)

The indicator of POADR would be:

$$POADR = \frac{Population\ older\ than\ the\ Old\ Age\ Threshold}{Population\ 15 - Old\ Age\ Threshold} x\ 100$$

In aging populations with increasing life expectancies, the POADR is manifestly lower than the OADR, as the prospective age increases in population projections. For example, in high life expectancy regions, in 2005, the prospective age of 15 years of remaining life expectancy was already 68.7 (compared to the 65 years old used in the OADR) and was expected to increase to 72.8 in 2045 projections.

Other population aging indicators have sought to include adjustments regarding the level of economic dependency. The level of production in a population depends on the ability of individuals to engage in market activities, their level of productivity, and the amount of capital involved. We are looking at them in more detail in the following section. You can find an inventory of many of these indicators in Spijker (2015), but new ones appear in the literature from time to time.

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2 Economic implications of Population Aging

2.1. Economic Indicators of Population Aging

If we want to assess economic sustainability, or the economic dependency of population aging, the use of pure demographic indicators may not be enough. This is why many researchers have proposed adjusted measures of population aging, considering the number of pensioners vs. workers, the relationship between non-workers over workers, etc.

In this sense, one of the most widely used approaches is the Economic Support Ratio, a measure developed within economic demography and closely linked to the concept of the Demographic Dividend.

2.2. The Economic Support Ratio and the Demographic Dividend

In the population aging process, there is a temporary stage in which population dependency diminishes, as the working-age population increases more rapidly than the other dependent groups. Several researchers have attached this low dependency period to increases in economic growth. Bloom & Williamson (1998) showed that part of the Asian economic miracle was explained by the faster growth of the working-age population compared to the rest from 1960 to 1995. They estimated that changes in the population age structure accounted for about one-third of the observed economic growth in Asia during that period. This is explained by the fact that a lower dependency ratio can free more resources for investment, and the increasing longevity might foster saving behaviors, creating positive effects on income. This is what has been called the demographic window of opportunities or the **demographic dividend** (Cutler et al., 1990).

Mason (2005) reformulated this concept by using an **Economic Support Ratio** (ESR) based on Labor Income and Consumption age profiles. To formalize this demographic dividend, the economic output per capita in a population can be expressed as the product of these two terms:

$$\frac{Y_t}{N_t} = \frac{L_t}{N_t} \cdot \frac{Y_t}{L_t} \qquad [1]$$

where Yt is the total output, Nt is the effective number of consumers, and Lt is the effective number of producers in year *t*. The effective number of producers is the population weighted by the income age profile, and the effective number of consumers is the population weighted by the consumption age profile. The support ratio is given by the ratio of effective producers (Nt) to the number of effective consumers (Lt). Both economic profiles are adjusted to their correspondent National Account identity, so the profiles are representative of the economy of the country.

Furthermore, by taking the natural log of both sides of Equation 1 and deriving it in respect to time, Mason et al. **(**2016**)** obtained the following disaggregated rate of growth:

$$\dot{y}_t = (\dot{L}_t - \dot{N}_t) + \dot{y}_t^l \qquad [2]$$

Hence, the rate of growth of the output per effective consumer ($\dot{y}_t$) equals the sum of two components. The first component, in parenthesis, is the difference between the growth in the number of effective producers and the growth in the number of effective consumers (i.e. the rate of growth of the Economic Support Ratio) and refers to the first dividend. With this component, we can easily see the positive impact that a faster growth rate of the working-age population (effective producers) compared to the lower growth rate of total population (effective consumers) can have on economic growth. In other words, if the growth rate of the Economic Support Ratio is positive, it means that producers are increasing faster than consumers, and, therefore, there can be a positive impact of population changes on income per capita in the population.

However, these researchers include a second component, that was named as the second dividend. This refers to the increase in the output per worker and the creation of wealth that arises in response to population aging, where capital-deepening can ultimately increase the output per effective consumer.

The First Dividend refers to a temporary effect, as producers will stop increasing faster than consumers as population aging advances. However, the Second Dividend can be a result of adequate investments due to the savings produced by the First Demographic Dividend. Unfortunately, the demographic dividends are not automatic and depend on institutions and policies to transform changes in population age structure into economic growth. The labor market needs to create enough opportunities for the growing working-age population, and a developed financial market is necessary to fulfill individuals' willingness to save (Mason, 2005).

Although the Demographic Dividend has finished or is already finished in many high- and middle-income countries, it is still a valuable framework to understand the impact of decreases in economic dependency ratios on the economy in many countries and regions (Baerlocher et al., 2019; Dramani and Oga, 2017).

2.3. Economic profiles and the Lifecycle Deficit

To obtain the economic profiles to estimate the Economic Support Ratio, and therefore, the demographic dividends presented by Mason (2005), Lee and Mason (2011) proposed a methodology to disaggregate economic profiles or, more specifically, National Accounts, by age. This methodology also allows us to assess the economic impact of changes in the population age structures and understand how resources flow from one age group to another at the national level. Their international project is called National Transfer Accounts, and you can find more information about the methods and applications at their webpage (www.ntaccounts.org). During the lifecycle, there are usually two periods where individuals are dependent on transfers from the working population, while they are children and older people. While in these two moments, individuals consume much more than they produce and need to finance this deficit by receiving transfers from the working-age population. The working-age population, therefore, needs to produce a surplus to transfer to the two dependent groups. These transfers can be private (done within the family), or public (through governments), but they can also be asset-based reallocations as savings or financial loans. This

Lifecycle Deficit (*Labor Income minus Consumption*) can be resumed in the following equivalence where it equals intergenerational transfers received (from other age groups) minus transfers given (to other age groups) and asset-based reallocations for each age group, as specified in:

$$LY_i - C_i = (TG_i^+ - TG_i^-) + (TF_i^+ - TF_i^-) + (AY_i - S_i) \quad [3]$$

The equation [1] must be held at all ages (i). There, LY refers to Labor Income, C to Consumption (private and public), TG to Government or Public Transfers, TF to private transfers, and AY to Asset Income and S to savings (which indicate together the net Asset-Based Reallocations). The + sign designates transfers received, while the – sign labels transfers given.

To estimate each component of the equivalence, the economic profiles are distributed by age and adjusted to National Accounts values. The steps to obtain this final profile are as follows. First, the average (or per capita) amount by age is calculated in gross values from a household income survey (or official records if these are available). Second, the aggregate value of the profile should be obtained to be fitted to the equivalent macro-control from the National Accounts. The aggregate value is calculated summing up the multiplied per capita value by the population age structure as following:

$$X = \sum_a \quad x(a) \cdot P(a) \quad [4]$$

where X is the aggregate value of the economic profile obtained through the per capita profiles (x) and the population (P) by age (a). Finally, each age value is adjusted to the representative of the equivalent macro-control from the National Accounts (X_{NA}) to obtain the definitive age profile, as:

$$x_{NA}(a) = x(a) \cdot \frac{X}{X_{NA}} \quad [5]$$

With the adjustment of profiles to the National Accounts, the Lifecycle deficit provides a more realistic view of the level of economic dependency for each group and, moreover, allows us to understand from which source each individual is dependent on (public, private, or financial).

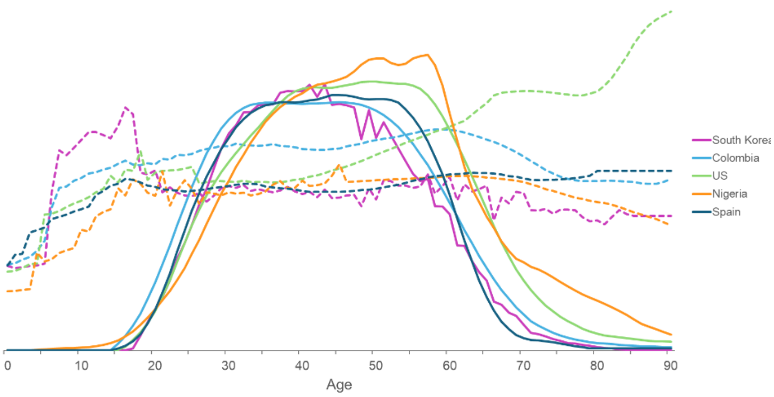
The application of National Transfer Accounts methods implies the utilization of diverse data. On one side, from household surveys or official records (like utilization of services or social transfers) to estimate the average profiles by age, and on the other, from National Accounts to adjust the profiles to the national level. The Lifecycle deficit is very useful for country comparisons and over time, of how much changes differences in age structures impact on the economy, but also to understand levels of dependency among different population age groups. First, it gives a harmonized method to obtain economic profiles. Secondly, the values are adjusted to National Accounts and more representative of the current economic level. Finally, it gives exact information about the moment that an age group produces enough labor income to finance their own consumption, and the moment where this surplus ends. The comparison of the economic profiles per capita of different countries, as seen in Figure 1, shows the persistence of the dependency age groups among countries (and over time), where children and older individuals experience economic dependency (their consumption is over their labor income). Nevertheless, some differences arise in consumption profiles. In this Figure, for example, South Korea shows a higher spending on education among young individuals, and the United States experiences a higher consumption in health for older individuals. Labor income profiles also show differences, where we can see Nigeria and the US standing out in the level of income of older individuals, as they extend their working lives longer than in other countries.

When looking at aggregate profiles of consumption and income, representing the real population, as in Figure 2, we can observe the importance of the population age structure. In this case, profiles differ greatly from the previous ones, and much more between countries. For example, in Nigeria, the consumption of young individuals who do not produce enough income to finance it is still very important due to their large share in the population. Moreover, consumption of middle-aged individuals in the US is quite high compared to the other countries, which might refer to the baby boom generation.

What changes greatly between countries is the way this economic dependency is financed. In Figure 3 there is a triangle comparing the relative importance of the three sources of income to finance consumption of older populations (65+) in different countries: Public transfers, Private transfers, and Asset-Based reallocations. While in Europe, the older population relies almost exclusively on public transfers, in Mexico or Philippines, they rely on Asset-Based income. Private transfers are positive more important in several Asian countries, but they are negative for many other countries, meaning that older ages tend to give more than receive private transfers.

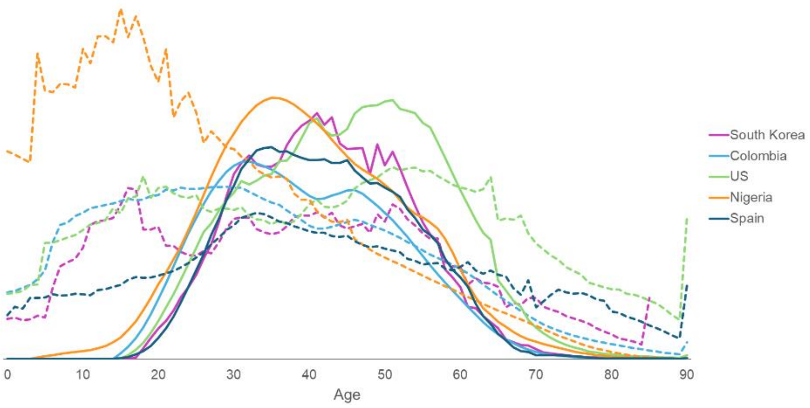
National Transfer Accounts have been collaborating closely with the United Nations, and they have published a manual that you can assess to know more specific information about the methods (UN, 2013).

Figure 1. Labor income and consumption age profiles among different countries, normalized values per capita



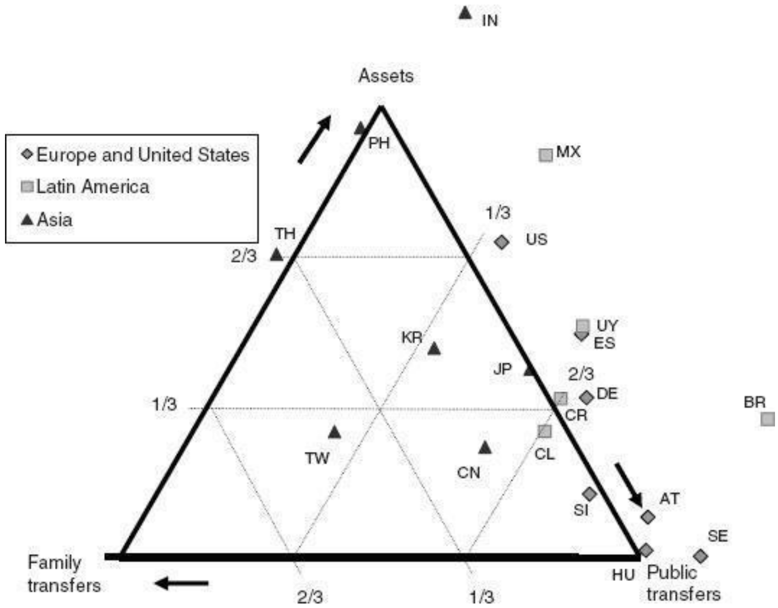
Source: National Transfer Accounts profiles normalized by Labor Income between ages 30 to 49. Labor income corresponds to continuous lines, and consumption to dashed lines.

Figure 2. Labor income and consumption age profiles among different countries, normalized population values



Source: National Transfer Accounts profiles normalized by Labor Income between ages 30 to 49. Labor income corresponds to continuous lines, and consumption to dashed lines.

Figure 3. Sources of financing the Lifecycle deficit of individuals aged 65+ among diverse countries.



Source: National Transfer Account profiles. The acronyms refer to: AT- Austria; BR – Brazil; CN – China; CR – Costa Rica; CL – Colombia; DE – Germany; ES – Spain; HU – Hungary; IN – India; JP – Japan; KR – South Korea; MX – Mexico; PH – Philippines; SE – Sweden; SI – Slovenia; TH – Thailand; TW – Taiwan; UY – Uruguay; US – United States

2.4. Extensions of the Demographic Dividend: Education, Gender and Health

One of the main criticisms of the NTA approach was the lack of heterogeneity in the economic profiles. NTA profiles only refer to averages of the whole population, and don't say much about social inequalities. The first change was to introduce gender differences. To fully account for the economic profiles of women, it was considered that the unpaid production should be included. We refer to all the productive activities done outside the market (unpaid) and related to domestic tasks and care. Within the National Time Transfer Accounts framework, similar economic profiles of production and consumption of housework and care were estimated and disaggregated by age (Donehower, 2019, Counting Women's Work). This methodology can be found in the recently published manual of the United Nations (UN, 2025). Moreover, NTA profiles have been also disaggregated by socioeconomic variables, mainly educational levels and socioeconomic variables, to account for these inequalities in economic profiles.

In both cases (gender and socioeconomic levels), the disaggregation allows to apply an extension of the demographic dividend that takes into account not only changes in population age structures, but structural changes related to increases in women's labor income (Oosthuizen and Lilenstein, 2018), and educational expansions (Rentería et al., 2016), for example. In the specific case of educational expansion, this has often occurred at the same time as the beginning of the population aging process. This has led some researchers to conclude that the demographic dividend was not so much demographic, but rather a consequence of the improvement in the population's educational levels (Lutz et al., 2008).

The study of how improvements in health at older ages can increase their labor supply and income and have an impact on the economic sustainability of population aging, has recently been approached (Matsukura et al., 2018; Scott, 2023; Fernandes and Queiroz, 2024). This has been called the silver or the longevity dividend. However, although it is true that these improvements imply that a more active population could be economically active for a longer period, their labor activity is strongly defined by employment laws and pension rules, and, therefore, it is subject to eventual changes in these regulations.

To conclude, it is important to understand that population aging and its impact on economic sustainability are a complex scenario that cannot be handled using only pure demographic indicators.

2.5. Other Aspects that Influence the Economic Sustainability of Population Aging

It is important to be aware that many factors influence the economic profiles used to measure economic dependency. Therefore, when looking into projections, different policies could impact the economic dependency. Among others, there are the labor activity rates, the pension systems, and the level of labor productivity, such as human capital (education and health). We can also consider the fact that labor income has an impact on asset-income that is not being explored in the economic profiles of the Lifecycle deficit. We are going to introduce some of these issues and how they can affect economic sustainability.

Regarding labor activity rates, gender gaps are one of the main components that impact labor income. First, many countries have experienced increases in the participation of women in the labor force, reducing the gap, although is still present. The lower activity of women is attached to their family responsibilities, and the way governments deal with the inactive periods due to them. For example, in Scandinavian countries, where mothers have the right to take long maternity leaves without disconnecting from the labor force, the activity rates of women are persistently higher than in the rest of Europe. Therefore, during the period when women are taking care of children, they don't lose much income and continue to be employed. In Central European countries, though, mothers reduce their hours of work, engaging in part-time jobs, which inevitably decreases their labor income. In the case of Southern European countries, mothers used to have a higher tendency to become unemployed or inactive. These differences between countries are mostly the consequence of government regulations related to employment, maternity leaves, childcare availability, and school hours' schedules (Hegewisch and Gronick, 2013).

Other aspects of labor activity that need to be acknowledged are unemployment rates, which vary greatly from one country to another, especially among younger and older workers. Policies that incentivize younger labor income may have a great impact on the labor income of younger individuals (Caliendo and Schmidl, 2016). In the case of middle- and lower-income countries, the level of formality of employment would be crucial, not only to define labor income, but also to understand the importance of this income to define the sustainability of future dependency stages in current workers.

Pension systems might not have a direct impact on the economic profile of the labor income, besides changes in the retirement age. However, they greatly define how economic dependency is financed at older ages. There are mainly two types of pension systems: Pay-as-you-go (PAYG) systems and Funded systems. PAYG systems are usually public and are financed by the current working population through contributions. There is no accumulation of pension wealth, and, therefore, it is a redistributive system. Examples of PAYG systems are in many European countries as Spain, France or Germany. Funded systems can be both private and public. In this case, individual workers rely on their own savings (if public, these can be mandatory, and if private not), that are further invested in financial markets to obtain greater returns. These systems can be also funded through employer's contributions. Examples of these funds at the public level are being applied in Chile, Ireland and the UK. Finally, pension transfers can also be non-contributory and financed through general taxes. These are usually given to individuals who have been inactive all through their lives and might respond to a necessity to relieve poverty. All these different schemes have varying consequences for intergenerational transfers and can be incentivized through tax reductions. Within a country, more of one scheme can co-exist or governments can be applying different schemes at the same time.

Another crucial aspect of labor income concerns labor productivity. In this sense, investments in human capital (health and education) are critical. When looking into population projections to measure economic sustainability, it is extremely important to consider these two aspects. Labor income inequalities by education level are evident, and, therefore, education expansion might bring increases in total labor income. Moreover, future projections need to consider that the levels of health are improving, as well as life expectancies. This implies that we are arriving healthier and more active at older ages, which allows extending, as well, our working lives. Many government reforms include extensions of retirement ages that would have a clear impact on labor income profiles.

Finally, there is another aspect of economic sustainability that has been less explored, which refers to asset income. Asset income is becoming a more important aspect of generational sustainability as it refers to income coming from accumulated wealth. In a context where wealth inequalities are increasing, and having a striking age component in this accumulation, it appears important to asset income jointly with labor income to assess economic generational sustainability. Loichinger et al. (2014) explore this dimension in their estimation of different dependency ratios, and, although population aging is undeniably raising dependency ratios with any indicator, accounting for asset-based income or asset-based reallocations implies a relief in this ratio.

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3

Health implications of Population Aging

3.1. Trends in Life Expectancy and Epidemiological Transition

3.1.1. Increased Longevity: A Recent Phenomenon

For most of human history, life expectancy was extremely low. In hunter-gatherer societies, it was estimated to be around 25 years, while in England in 1700, it barely reached 37 years (Cutler, Deaton & Lleras-Muney, 2006). This scenario began to change significantly from the 18th century in more developed countries, culminating in the 20th century with an unprecedented decline in mortality, leading to an increase of approximately 30 years in life expectancy.

The main causes of this remarkable extension are:

- Nutritional improvements: Better diets and access to food.

- Advances in public health: Large-scale public works projects like sanitation, water supply, and swamp drainage, alongside public-sector-driven health initiatives.
- Vaccination and medical treatments: Discoveries and widespread dissemination of vaccines and antibiotics, especially from the 20th century onwards.
- Urbanization: Though complex, with initial negative impacts, urbanization progressively raised living standards and increased access to health services.
- Early life care: Long-term effects of care received early in life.
- Income: Increased income improves mortality by enhancing access to essential resources like quality healthcare, nutrition, and safer living environments, and by facilitating investments in public health infrastructure.

While vaccination and medical treatments were important, a significant portion of the reduction in deaths from infectious diseases occurred before major medical discoveries (Cutler, Deaton & Lleras-Muney, 2006; McKeown, 1976). This reinforces the fundamental role of nutritional improvements and public health in the decline of mortality in developed countries.

Moreover, urbanization played a dual role. While it can be positive for health by raising living standards and increasing access to services, in the initial phases of the European process, larger and denser cities face unsanitary conditions and a greater spread of communicable diseases. In less developed countries, the lack of sanitation in urban agglomerations still leads to a higher prevalence of infectious diseases, especially among children, underscoring the need for improved living conditions and urban reforms.

3.1.2. Income, Health, and Global Disparities

Cutler, Deaton, and Lleras-Muney (2006) minimize the direct role of income in the decline of mortality, arguing that both health improvements and increases in income result from new ideas and technologies. For them, a country's health stems from its institutional capacity and political will to implement known technologies, rather than being an automatic consequence of increased income.

In less developed countries, life expectancy is still lower compared to wealthier nations. However, in the years immediately following World War II, global differentials in life expectancy decreased. This happened because progress against mortality observed in more developed countries was quickly adopted globally, as seen with the spread of water supply systems, vaccines, and antibiotics.

Within countries, people with different incomes, education levels, habits, and socioeconomic statuses also show distinct mortality patterns. For the authors, the explanation for the relationship between income and health lies in access to health services. Furthermore, education, especially for women, is highlighted as crucial for improving primary childcare, contributing to a reduction in mortality at younger ages.

3.1.3. Epidemiological Transition: Changes in Disease and Death Patterns

The concept of Epidemiological Transition, proposed by Omran (1971), describes the progressive evolution of mortality, morbidity, and disability patterns in a specific population, alongside other demographic, social, and economic transformations (Prata, 1992; Horiuchi, 1997; Schramm et al., 2004). Horiuchi (1997) expands this idea into a five-stage model of the epidemiological transition of human mortality:

Stage 1: Reduction in mortality from external causes: Characteristic of hunter-gatherer societies, where accidents, homicides, and attacks were predominant. With agricultural societies, these deaths became increasingly rare.

Stage 2: Reduction in mortality from infectious, maternal, and perinatal diseases, and nutritional deficiencies. This stage began in more developed countries in the 18th century, leading to the greatest gains in longevity. Factors such as increased population size, density, and contact between communities, as well as nutritional deficits and unsanitary urbanization, initially aggravated the proliferation of infectious diseases. However, with improved conditions, mortality declined, first in children and then in adults (especially from tuberculosis). This decline was due to both a reduction in mortality crises and a regular decrease over the years, driven by improvements in living standards, public health, and personal hygiene (Cutler et al., 2006; Soares, 2007). However, this period also saw an increase in the prevalence of degenerative diseases at older ages.

Stage 3: Decline in deaths from cardiovascular diseases occurred from the mid-20th century onwards, with advances in curative medicine, health systems development, and preventive care (i.e. diabetes and hypertension), primarily benefiting older age groups.

Stage 4: Refers to a decline in cancer mortality. Nevertheless, a transition to this realm is still expected soon.

Stage 5: Senescene slowing down is already occurring. But again, another future transition, with substantial changes could still be observed in comming decades.

Despite the heuristic value of these stages, it is important to note that the epidemiological transition does not always occur uniformly. In less developed countries, for instance, there's still a significant coexistence of chronic degenerative diseases (characteristic of advanced stages of the transition) and infectious diseases (characteristic of initial stages). This overlap creates a dual challenge for healthcare systems. Furthermore, even within the same country or region, significant socioeconomic differentials can exist in the epidemiological transition process, reflecting the morbimortality profile of different social strata, given disparities in access to information, health services, nutrition, and housing.

3.1.4. Reverse Transitions and Future Challenges

Horiuchi (1997) highlights the concept of reverse transitions, which refer to an increase in mortality levels in a region or group, associated with factors such as:

- Precarious working conditions, inadequate food, housing, and sanitation in the initial stages of industrialization.
- Adoption of less healthy lifestyles (alcohol consumption, high-fat diets, smoking).
- Emergence of new infectious diseases (e.g., HIV/AIDS, Covid-19).
- Pollution and environmental degradation.
- Social alienation (depression, homicides, suicides).

3.1.5. Critiques to the Classical Theory of Epidemiological Transition

The classical theory of Epidemiological Transition formulated by Omran (1971) posits a linear and progressive shift in mortality patterns, from infectious to chronic diseases, as societies develop. However, this view has been widely criticized for its simplicity and failure to capture the complexity of global realities. Olshansky and Ault (1986) were pioneers in introducing the concept of a "fourth stage," the "age of delayed degenerative diseases," suggesting that longevity gains are not limited to a simple replacement of diseases but can involve the postponement of morbidity and the compression of illness into older ages. This perspective already pointed to a more dynamic transition with the potential to continue evolving beyond the initially proposed stages. Additionally, by analyzing the convergence and divergence in mortality, Vallin and Meslé (2004), show that health trajectories vary significantly among nations, with different "waves" of mortality decline impacting populations unequally. For the specific case of Latin America, Frenk et al. (1991) state that the epidemiological transition process in this region follows a "polarized-prolonged model." This means that there is an overlap of high incidences of pre-transitional stage diseases—infectious diseases and external causes—and post-transitional diseases—chronic illnesses—without a clear resolution of each stage of the transition process.

3.2. Rectangularization of Survival Curves

As mortality rates have declined across all age groups, particularly among younger populations, the distribution of deaths has shifted significantly. This phenomenon is known as the *rectangularization* of survival curve or compression of the mortality curve. Historically, mortality was high at younger ages, leading to a diagonal-shaped survival curve. With increased longevity, a larger proportion of the population survives to much older ages, resulting in a curve that becomes increasingly "rectangular" – indicating that most deaths occur within a narrow range of advanced ages.

3.3. The Relationship Between Morbidity and Mortality

Morbidity, in common usage, is a general term for the absence of health, and disability is the most frequently used metric for estimating morbidity. Morbidity itself is an imprecise term often defined in different ways, usually denoting impaired health of some kind other than death. There is a clear dynamic between changes over time in morbidity and in mortality, since fatal and nonfatal outcomes are generally correlated. However, according to Fries (2012), simply postponing morbidity (delaying when illnesses begin) would improve overall health. In contrast, if we only postpone mortality (delaying death) without also delaying illness, people will likely experience more years of poor health. Fries also emphasize that accurately predicting future health trends depends on the dynamic interaction between morbidity and mortality trends. If people live longer but get sick earlier, their lifetime experience with illness will increase. Conversely, if the onset of illness is delayed more significantly than death, the total years spent in poor health over a lifetime would likely decrease.

The Morbidity Compression Paradigm, proposed by James Fries in 1980, hypothesizes that as life expectancy increases, the onset of significant chronic diseases and disabilities can be postponed to a greater extent than the age of death. This means that the period of life spent in poor health or with significant disability would be "compressed" into a shorter time frame at the very end of life. The core idea is that by promoting healthier lifestyles and effective preventive measures, individuals can maintain good health for a longer portion of their lives. This will lead to a more "rectangular" survival curve where people live longer and healthier, with a relatively rapid decline just before death, rather than experiencing prolonged periods of illness.

3.4. Measures of Quality-adjusted Life Expectancy

While traditional life expectancy measures the average duration of life in a population, a more comprehensive understanding of population health and well-being requires moving beyond mere longevity to incorporate the quality of life lived during those years. Several approaches exist to estimate quality-adjusted life expectancy, providing richer insights into a population's health status and overall well-being.

● **Happy Life Expectancy:** This metric integrates information on individual happiness and life satisfaction to complement life expectancy. The idea is to capture not only how many years a person lives, but how many of those years are lived in a perceived state of happiness or contentment. It's an important measure for assessing subjective well-being at a population level, recognizing that health isn't just the absence of disease, but also a state of mental and emotional well-being.

● **Healthy Life Expectancy (HALE):** This measure uses information on health status to estimate the number of years a person can expect to live in good health, free from severe illnesses or significant limitations. Unlike crude life expectancy, HALE discounts years lived with chronic diseases, functional limitations, or disabilities. By focusing on specific aspects of health, this metric can be detailed in several ways, considering:

I. Chronic diseases: The prevalence and impact of conditions such as diabetes, hypertension, heart disease, etc.
II. Limitations and disabilities: Physical, mental, sensory, or cognitive impairments limit a person’s ability to perform social or professional activities often long-lasting.

III. Basic Activities of Daily Living (ADLs): The ability to perform essential tasks like eating, dressing, bathing, and moving around.

3.5. Promoting Healthy and Active Aging

3.5.1. The Relevance of Population Aging and Its New Paradigms

Population aging is an undeniable global phenomenon. In 2019, there were approximately 700 million older adults worldwide, a number expected to double by 2050, with the population over 80 years old tripling in the same period (Dogra et al., 2022). This faster growth is occurring in regions such as East and Southeast Asia, Latin America, and the Caribbean, highlighting the urgent need to address aging strategically and comprehensively.

Longevity was traditionally associated with an inevitable increase in morbidity. However, starting in the 1980s, a new paradigm emerged in gerontology, based on the idea of compression of morbidity (Fernández-Ballesteros et al., 2013; Fries, 2012). Research indicates that since the 1950s, mortality after age 80 has consistently decreased, and human senescence has been delayed by a decade, strongly associated with optimized behavioral and health practices (Christensen et al., 2009). This new understanding emphasizes the plasticity of the human being throughout life and into old age, with the possibility of health improvements even with increased longevity (Fries, 2012).

3.5.2. Redefining Healthy Aging: From the "Absence of Disease" to "Functional Ability"

The World Health Organization (WHO) has been instrumental in reshaping the concept of aging. In 2015, in the preface to the first World Report on Aging and Health, Margaret Chan, then WHO Director-General, clearly stated that "healthy aging is more than just the absence of disease" (Michel & Sadana, 2017). The new definition of Healthy Aging proposed by the WHO, and unanimously endorsed by all Member States in 2016, is "the process of developing and maintaining functional ability that enables well-being in older age" (Michel & Sadana, 2017; Dogra et al., 2022). This definition recognizes the continuous interaction between an individual and their environment, highlighting two crucial elements:

● **Intrinsic Capacity:** Refers to the combination of all an individual's physical and mental capacities. It results from a complex interaction between genetic inheritance, personal and health characteristics (including age-related physiological changes, accumulation of risk factors, and the impact of acute or chronic clinical conditions), as well as intellectual potential and psychological resilience throughout the life course (Michel & Sadana, 2017; Dogra et al., 2022).

● **Environment:** The environment in which an individual lives is dynamic and profoundly influenced by political, economic, and social norms, values, and resources. This includes the extent to which societies promote equal opportunities, prevent inequities, combat ageism, and ensure access to affordable health and social systems. The interaction between intrinsic capacity and environmental characteristics (including support and the physical surroundings) determines functional ability (Michel & Sadana, 2017; Dogra et al., 2022).

For the WHO, healthy aging is a reality possible for all and requires a shift in focus: from the mere absence of disease to fostering the functional ability that enables older people to be and to do what they value (Dogra et al., 2022).

3.5.3. Active Aging: A Multidimensional and Comprehensive Concept

The term Active Aging is a comprehensive concept that encompasses the idea of "aging well", connecting to other terms such as successful, productive, competent, and healthy aging (Fernández-Ballesteros et al., 2013; Caprara et al., 2013; Fries, 2012). Although there is no universally accepted empirical definition, there is a consensus that active aging embraces multiple domains, such as:

- Low probability of illness and disability.
- High physical fitness.
- High cognitive functioning.
- Positive mood and ability to cope with stress.
- Engagement with life (Fernández-Ballesteros et al., 2013; Caprara et al., 2013).

The WHO has identified six main determinants of active aging at the population level: behavioral styles, personal biological and psychological conditions, health and social services, physical environment, and social and economic factors (Fernández-Ballesteros et al., 2013). These determinants highlight that the responsibility for active aging is shared, requiring long-term public health programs and social policies (Caprara et al., 2013).

It's important to note that while Healthy Aging and Active Aging are often used interchangeably, there are nuances. Fries (2012) suggests that "healthy aging" might be a preferable term as it includes the improvement of physical, mental, and social health, whereas "active" may be more focused on the physical component. Nevertheless, the common thread running through all these concepts is the promotion of a good quality of life in old age, focusing not only on the absence of diseases but on the ability to maintain functioning and well-being.

Healthy and Active aging are, therefore, a multisectoral issue affecting all parts of society, including the environment, labor, gender issues, health equity, social participation, digital technology, education, and recreation (Dogra et al., 2022). This implies that public health initiatives should go beyond the prevention and management of specific chronic diseases, adopting a broader and more multidisciplinary approach. Focusing on domains such as physical function, cognitive function, mental health, social health, and sleep quality, that can create more inclusive and relevant public health research and messages that enhance active aging (Dogra et al., 2022).

3.6. Future Challenges for Health Systems in the Face of Population Aging

The unprecedented global demographic shift towards an older population supposes complex and multifaceted challenges for health systems worldwide. While increased longevity is an achievement of societies, it simultaneously demands a profound re-evaluation and adaptation of healthcare systems, of their structures, founding, and service delivery. The primary challenges that health systems face in this aging era include:

● **Climate Change:** The escalating impacts of climate change, such as extreme heat events, air pollution, and altered disease vectors, disproportionately affect older adults, who are more vulnerable to heat stroke, respiratory illnesses, and infectious diseases. Health systems will face increased pressure to manage climate-related health crises, develop resilient infrastructure, and adapt public health strategies to protect a growing older population.

● **Unhealthy Lifestyle Habits:** Despite advances in medicine, the prevalence of unhealthy lifestyle habits (e.g., sedentary behavior, poor diet, smoking, excessive alcohol consumption) contributes significantly to the burden of chronic diseases. As populations age, the cumulative effects of these habits manifest as a higher incidence of non-communicable diseases, necessitating more extensive and prolonged care, and challenging the sustainability of health systems.

● **Mental Health and its Effects on Chronic Illnesses:** Mental health issues, including depression, anxiety, and neurodegenerative conditions like dementia, are increasingly prevalent in older adults and often coexist with chronic physical illnesses. This comorbidity complicates diagnosis and treatment, increases healthcare utilization, and reduces overall quality of life. Health systems must integrate mental health services more effectively into primary and specialized care to address the holistic needs of an aging population.

● **Socioeconomic Inequalities:** Population aging often exacerbates existing socioeconomic inequalities in health. Disparities in income, education, and access to quality housing and nutritious food translate into significant differences in health outcomes and access to care among older adults. Health systems face the challenge of designing and implementing equitable services that reach vulnerable older populations, preventing further marginalization and ensuring universal access to healthcare.

● **Incorporation of New Medical Technologies:** The rapid development of new medical technologies, from advanced diagnostics and personalized medicine to innovative surgical techniques and digital health solutions, offers immense potential to improve the lives of older adults. However, health systems grapple with the ethical considerations, regulatory complexities, and substantial financial investment required for their widespread and equitable incorporation. Ensuring access to beneficial technologies without widening disparities or bankrupting health systems is a critical balancing act.

● **Increase in Healthcare Spending:** The aging demographic, coupled with a rising prevalence of chronic morbidities and the integration of technological innovations in the healthcare sector, significantly escalates healthcare expenditures. Therefore, ensuring the financial sustainability of healthcare systems, alongside maintaining their quality and accessibility, emerges as a strategic imperative. The central apprehension lies in the potential unsustainability of public healthcare spending driven by population aging under current financing models (Howdon & Rice, 2018). This scenario presents a complex challenge for policymakers, who must reconcile the provision of adequate care to an aging demographic with the fiscal viability of healthcare systems. The issue extends beyond sheer population volume, encompassing the intensity and long-term care requirements, with direct implications for resource allocation and service responsiveness.

● **Shortage of Qualified Labor** (Nurses, Doctors, etc.): An aging population not only increases the demand for healthcare services but also contributes to the aging of the healthcare workforce itself. Many healthcare professionals are nearing retirement, while the pipeline of new recruits, particularly in specialized geriatric care, struggles to keep pace. This creates a critical shortage of qualified nurses, doctors, and other allied health professionals, threatening the capacity of health systems to provide adequate care to their expanding elderly cohorts.

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4

Social Implications of Population Aging

4.1. Aging and Care

A very important aspect of population aging refers to the expected increasing demand of care. As explained in Section 3, a higher share of older adults translates into more individuals experiencing chronic illnesses, disability, and cognitive decline, which increases their need for support with daily activities, medical care, and emotional well-being.

This is a challenge for health and social care systems, labor markets, families and for society in general. In many contexts, care has traditionally been provided informally by women in the family, often without compensation or formal recognition. However, changing family structures, increased female labor force participation, and geographic mobility are putting pressure on this model.

4.2. Defining Care

The first challenge when tackling with care arises from its definition. What is care? Feminist theories were precursors in referring to social care as an important dimension of economies. In the 70s care was defined as part of the reproductive labor, usually done by women within families, needed to maintain the labor force in charge of the market production. Partly inspired by this previous notion, Fisher and Tronto (1990) defined care as:

"(…) everything that we do to maintain, continue, and repair our 'world' so that we can live in it as well as possible"

This comprehensive definition understands care as a social practice embedded on our lives; it emphasizes that we as individuals need care while can care for others. Despite its potential, it is a challenge to operate it for social research. In general, a restricted definition of care refers to activities involved in looking after others, particularly when they need it. It implies very different activities, and defining which ones are included will vary depending on the objective of the study. They can refer to:

- Direct physical care: e.g. bathing, dressing, feeding.
- Household work: e.g. cooking, cleaning, shopping.
- Medical care: e.g. administering medications, managing doctor appointments.
- Emotional and relational support: e.g. companionship, reassurance.
- Instrumental help: e.g. helping with phone use or internet, managing finances.
- Mental load of care: e.g. planning care, managing care needs, anticipating them.

The use of the term care is not universal, and many times researchers call it help or support. It is true that care might imply deep responsibility and emotional involvement, but more specific tasks defined by help and support can also be considered as care in many studies. One of the main challenges of studying care has been to make visible what has been previously taken for granted, giving relevance to the unpaid work that maintains populations. This is related to the difficulty of having a specific definition of care, because its meaning can vary for specific subgroups due to social aspects like gender, cultural differences or socioeconomic status.

However, in general care is described by the way in which it is arranged or provided, differentiating between formal or informal care. Formal care is usually paid, regulated and provided by professionals, while informal care is provided by family, friends or neighbors. Additionally, there are different scenarios in which informal care is provided, according to familiar decisions (e.g. paying privately for this undeclared work) or through governmental programs (e.g. economic transfers, care training and support) which aim to compensate informal caregivers for their work. While formal care is easier to measure, as it is attached to specific hours and prices, informal and unpaid care is more difficult to capture, as it lacks official records, as a formal contract. Moreover, many informal caregivers, don't consider what they do as "caring" and may not report it, because it is part of their routine tasks and something that they are expected to do, given pressing normal values.

Additionally, informal care often overlaps with other daily activities, making difficult to estimation of the time spent on it.

4.3 Care Provision

4.3.1 Measuring Unpaid and Informal Care

Since such activities fall outside the formal market, data collection on care providers, care activities and care intensity relies on surveys on how people spend their time. Two primary approaches are employed to gather this information:

● **Time Diary Method:** The first and most widely regarded approach, often considered the benchmark for time use data (Juster and Stafford, 1991; Van den Berg and Spauwen, 2006). This method involves participants recording all their activities, including caregiving, in a time diary over a defined period—usually a single day. Numerous countries conduct national Time Use Surveys (TUS) following this approach. Within Europe, the Harmonised European Time Use Survey (HETUS), coordinated by Eurostat, represents a notable example, with two completed rounds (2000 and 2010) and a third initiated in 2020.

● **Recall Method:** This second approach gathers information on time use (including care) based on stylized questions or activity lists. In this case, respondents answer specific questions about the time they have devoted to a list of tasks in a reference period (as the previous day, week or month). There are different macro-surveys (at the national or cross-national level) providing results on time devoted to non-market activities (including care provision) based on this recall method. At the European level, the most broadly used is the Survey of Health, Ageing and Retirement in Europe (SHARE).

The two methods of collecting data on unpaid care activities have their pros and cons, which need to be accurately considered according to the purpose of the specific analysis to be performed. In general, information collected by surveys based on time diaries is seen as more reliable, but less rich than that derived from a well-targeted set of stylized questions. Moreover, time diaries are much more expensive and difficult to collect. At the empirical level, surveys based on time diaries have been widely used to measure care, although there are critics about how adult care is included (Miranda, 2011; Francavilla et al. 2013; Vargha et al., 2017). Different national surveys based on the recall method have also been broadly employed to analyze intergenerational transfers of care. At the European level, SHARE underscores it as the most used dataset to investigate this type of transfer.

4.3.2. The Value of Care

To highlight the importance that (unpaid) care has in society and make its contribution visible, it has been necessary to value care in economic terms. This is extremely important in the case of women, who disproportionally participate in all these productive activities that have been taken for granted. By giving it an economic value, it can be highlighted, for example, its share of the Gross Domestic Product. This is crucial to understand gender inequalities and the importance of considering women's unpaid work as part of the production of a country. However, as it is an activity that is not mediated by a salary or monetary exchange, it is complex to give it value. There are several methods to value unpaid production that can be applied to care, but none of them is perfect:

● **Replacement Cost Method**: This is the most used one. It consists in valuing care as the cost to replace it with paid services. Depending on the activity of care, whether it is general or specialized, it will be more or less valued.

● **Opportunity Cost Method:** It assumes that the caregiver's unpaid work should be valued as per the income this caregiver is losing for not working in the labor market. Therefore, if a caregiver used to be an engineer, the cost of care should be valued as per its salary before assuming the care tasks. This type of valuing method is very useful to understand the loss of market production due to care demand, for example.

● **Willingness to Pay for Doing the Care Activity**: To use this method, it is necessary to ask individuals how much they would need to be compensated to provide care.

4.4. Care Needs

The relevance of care can also be studied by emphasizing individuals and populations care needs. These have usually been operationalized as individuals' limitations in performing Basic or Instrumental Activities of Daily Living (ADL or IADL). Different studies have used diverse activities to assess care needs. Moreover, there are some cases in which care needs are assumed when someone is facing limitations or needing help/support to perform any of these activities. Measuring care needs is relevant for understanding current and future care demands, and how policies and governments can respond to these.

A very relevant aspect of care needs regarding aging populations and the preparedness of societies to face the growing demand for care refers to the study of unmet care needs. These are very sensitive measures of how populations can support older individuals facing care needs. There are two main approaches to estimate unmet care needs (Vlachantoni et al.,2011):

● **Absolute Approach**: Based on those who report needing care but are not receiving help/support from anybody to deal with this need.

● **Relative Approach**: In this case, unmet care needs are based on individuals' perspectives of whether the help/support that they receive to address their needs is sufficient.

These measures are complementary but depend on specific questions that are not always included in population-based surveys. Moreover, they both depend on individuals' perceptions of the meaning of care/help and support that might vary due to gender, cultural values and other social categories. By studying care needs, we understand the other side of the coin, that of the care receivers. Recent research has shown that even though social care demand is expected to increase, the ways in which this will occur can vary depending on health systems' performance and success with active aging. Finally, unmet care needs are gaining attention within a novel area of study referring to care poverty, which emphasizes that social care needs are shaped by social inequalities and social care systems' gaps when these needs are considered.

4.4.1. Care quality

In line with discussions about unmet care needs is the issue of the quality of care. Given the previously mentioned differences between informal and formal care provision, assessing the quality of the care that is being provided or received is challenging, especially when the type of support being provided is not qualified or is given outside labor market regulations. From a demographic approach, there is scarce evidence assessing the quality of care. However, recent qualitative and epidemiological research has emphasized the relevance of this issue when designing, planning and implementing social care policies, especially within scenarios that are prone to meet care demands by mainly informal care provision.

4.5. Social Care Regimes

The way in which care demand and supply is organized varies according to specific social care regimes (also named as social care systems). This mainly refers to the different arrangements through which informal and formal caregivers meet individuals demands as well as to the different relationships established between states, markets and families for doing so. Social care regimes, theoretical and empirical typologies have been proposed as explanatory tools to understand the different ways in which care is being provided and usually do so by disentangling the degree of state and family involvement in caring. Previous typologies were mainly based on welfare states configuration (Pfau-Effinger 2005), but more recent typologies are using data on Long-term care service availability, and specific policies aiming to compensate family members providing care (van Damme, et al., 2025).

4.6. Debate on the Availability of Caregivers for Older Individuals

4.6.1. Options for Long-Term Care

Long-term care encompasses the continuous assistance required by individuals who, due to chronic illness, disabilities, or the aging process, face limitations in performing basic daily activities. These services, whether provided in institutions or within the community, are fundamental to the well-being of a wide range of people, regardless of their age.

Notably, non-institutional services, also known as personal assistance, personal care, or home and community-based services, empower many people with disabilities to maintain their independence, avoid institutionalization, and actively participate in family, community, and economic life. This assistance can be formally purchased or obtained from family, friends, and other volunteer helpers. Despite ongoing efforts to improve the long-term care system and reduce spending, a lack of detailed information regarding costs, specific services, care settings, and the extent of unpaid help received continues to limit its progress.

Over the past decades, several European countries have developed strategies to strengthen community-based services, aiming to replace traditional care models. Deinstitutionalization, defined as the development of community-based services as an alternative to institutional care settings, has become the hallmark strategy for individuals with limited autonomy across Europe (Ilinca et al., 2015). This shift is driven by two core arguments: prioritizing users' quality of life and increasing the sustainability of care systems through approaches based on solidarity.

The first argument is rooted in the belief that the protection of fundamental human rights must be central to all care systems, as community-segregated institutions can deny residents many basic rights, whereas an independent life within the community promotes them. The second argument is supported by growing evidence that community-based care solutions can achieve better outcomes for users and their families at lower or comparable costs relative to institutional care (Economic Policy Committee, 2009). Furthermore, community-based care better aligns with the preferences of Europeans, the majority of whom express a strong preference for aging in place, regardless of their nationality or cultural background (Economic Policy Committee, 2009).

4.6.2. Availability of Caregivers

As the care needs of the growing older population are increasingly met by family and friends, governments worldwide are recognizing the critical importance of supporting these informal caregivers and safeguarding their health and well-being. By gaining a deeper understanding of the factors influencing informal caregivers' relationships and their experiences with formal support systems, we can better discern the linkages between these two forms of care. This knowledge is paramount for improving and increasing the receipt of appropriate formal care by family caregivers and, consequently, for the frail or disabled older adults themselves.

Demographic supply-side factors of caregivers are like those of economic sustainability challenges. As populations age, the proportion of potential care recipients increases, while the ratio of working-age adults to seniors often decreases, limiting the pool of available caregivers, especially informal (kinship-related) ones. Lower birth rates in previous generations mean fewer adult children are available to care for aging parents. Socioeconomic factors also play a significant role. Historically, and still largely, women disproportionately assume informal caregiving roles. Societal shifts in women's participation in the workforce and their own aging can affect their availability as care providers. The willingness and ability of individuals to take on caregiving roles, whether formal (paid) or informal (unpaid), is influenced by their employment status, financial needs, and career aspirations. The availability of financial compensation or support for caregivers—such as paid family leave or caregiver stipends—can significantly impact willingness to provide care, especially for informal caregivers. For formal caregivers, the availability of appropriate training and certification programs, as well as the cost and accessibility of such programs, affects the supply of qualified professionals.

Personal and family dynamics are equally important. Smaller family sizes, geographic dispersion of family members, and changing family norms can reduce the number of available informal caregivers. The physical, emotional, and financial strain experienced by caregivers—such as burnout, stress, and health decline—can lead to their withdrawal from caregiving or limiting their capacity, thus reducing overall availability. Individuals with their own chronic health conditions or disabilities may be less prone to providing care. Additionally, societal and cultural expectations regarding family responsibility for older adults' care can influence the willingness of individuals to take on caregiving roles.

4.6.3. Impact on the Health of Caregivers

Kinship ties often provide invaluable social, emotional, and instrumental support in times of need, acting as an intrafamilial safety net that facilitates resource exchange and offer protection against various adverse health and well-being outcomes (Carr & Utz, 2020; Freedman et al., 2024). However, this role comes at a significant cost.

Numerous studies consistently show that this caregiving function has substantial negative impacts, especially on female caregivers' health and well-being. This demographic group frequently experiences a deteriorated general health status, a higher prevalence of chronic conditions like angina, arthritis, and hypertension, and a reduced perception of their quality of life. Psychological issues such as anxiety and depression are also more frequently experienced by intense caregivers. Despite often being older and already managing chronic health problems that can be exacerbated by caregiving, many persist for extended periods in conditions detrimental to their own health, with research indicating higher rates of morbidity and mortality compared to the general population (Martinez-Marcos & De la Cuesta-Benjumea, 2024).

4.6.4. Alternatives for Care Provision

Facing the growing challenge of caregiver scarcity in the future, as well as rising costs of long-term care solutions with shrinking contributors to the welfare system, there's significant uncertainty regarding solutions, especially at governmental levels. However, literature already points to alternative practices that can mitigate the effects of this shortage, such as cohousing and care robots.

Cohousing is emerging as a promising alternative to nursing homes and elderly residences. It's a form of group living that clusters individual homes around a "common house"— a shared space and amenities. This model is entirely run and controlled by the members themselves, based on mutual support, self-governance, and active participation. Physically, cohousing is designed to promote easy social interaction among its members, typically featuring a common house for shared meals and events. Two main cohousing models exist: the intergenerational (or family-based) and senior cohousing, the latter intended for age-peer groups over the age of fifty or so. Cohousing allows for living "apart and together" with a collaborative group of neighbors who know each other and share certain values, working to develop the social capital that creates and maintains a sense of community.

According to Brenton (2013), senior cohousing emerges as a significant investment by older people themselves in social capital and mutual support. It offers an additional option for informal care and housing needs as people age, contributing to keeping them active, healthy, and engaged, and consequently, reducing the demand for health and social care services. Essentially, this model provides an ideal blend of privacy and communality, allowing older adults to maintain their autonomy while enjoying the benefits of a supportive community.

Over the past decades, numerous robots have been developed to assist older individuals in various activities, a crucial advancement given the increasing imbalance between the number of individuals needing care and the decreasing number of caregivers. Care robots are viewed as a promising technological development to mitigate this disparity. These robots, which can be semi-independent or fully independent, support caregivers and/or older adults in physically assistive tasks. Many studies have explored the use of robots in aged-care settings, their effectiveness, factors influencing older adults' acceptance or rejection, and their attitudes toward socially assistive robots (Bedaf et al., 2015; Vandemeulebroucke et al., 2018). However, as robot technology advances and the conviction of its use in aged-care practices builds, there is a growing need for ethical reflection on its implementation.

Although many robots have been developed, only a small percentage are commercially available, and development seems to be guided by technical ambitions. To prolong independent living, physical support is an inevitable and necessary step. Nevertheless, it will be a long time before a robot is capable of physically supporting multiple activities in an elderly person's home to enhance their independent living. Existing assistive technologies, such as stair lifts, patient hoists, and home accessibility adaptations, have come a long way in supporting the independence of the elderly (Bedaf et al., 2015). Care robotics is an emerging field within assistive technologies, with many developments underway, and robots for the elderly continue to be developed to support a range of activities.

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